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MIL-A-62565A(AT)
15 April 1994
SUPERSEDING
(see 6.6)

MILITARY SPECIFICATION

AIR CLEANER, ENGINE

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, test, manufacture and acceptance requirements for the engine air cleaner, shown on Drawing 19207-12307204, referred to herein as the air cleaner (see 6.5.1), and the replaceable air filter element, shown on Drawing 19207-12329100, referred to herein as the filter element (see 6.5.2). Unless indicated otherwise, references to the air cleaner can be understood to include the filter element (see 4.1.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2940

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SPECIFICATIONS

FEDERAL

GG-P-455	Plates and Foils (Photosensitive Anodized Aluminum).
P-D-220	- Detergent, General Purpose.
QQ-P-416	Plating, Cadmium (Electrodeposited).

MILITARY

MIL-P-514	- Plate, Identification, Instruction, and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-81O	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MS 33660	- Tubing End, Hose Connection, Standard Dimensions for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

19207-12307204	- Air Cleaner, Engine

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19207-12329100 Element, Air Filter

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

STANDARDS

SAE J726 - Air Cleaner Test Code, dated June 1987.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

ASTM-B633 - Zinc on Iron and Steel, Electro-deposited Coatings of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. Materials used shall enable the air cleaner and filter element to meet the performance and environmental requirements of this specification.

3.2.1 Dissimilar metals. The air cleaner and filter element shall be designed and fabricated to minimize galvanic corrosion in dissimilar metal couples as specified in MIL-STD-889.

3.2.2 Standard parts. Military standard parts shall be used wherever possible.

3.3 Design and construction. The design and construction of the air cleaner and filter element shall be as specified herein and in accordance with Drawings 19207-12307204 and 19207-12329100 respectively.

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3.3.1 Air cleaner.

3.3.1.1 Weight. Weight of the air cleaner (including a new filter element) shall be not greater than 100 pounds.

3.3.1.2 Dimensions. Dimensions of the air cleaner shall be in accordance with Drawing 19207-12307204.

3.3.1.3 Tubular joints. The tubular ends of the air cleaner inlet and outlet and the scavenging outlet shall have externally beaded ends conforming to MS 33660, type A, except the bead height tolerance shall be + 0.003, - 0.010 inch.

3.3.2 Filter element. The filter element shall be of a one-piece design, firmly bolted or clamped in lace and easily replaced using hand clamps or common hand tools (see also 6.5.2).

3.3.2.1 Weight. The weight of a new filter element shall not exceed 30 pounds.

3.3.2.2 Dimensions. Dimensions of the filter element shall be in accordance with Drawing 19207-12329100.

3.3.3 Protective coating. A protective finish per class 1A, MIL-C-5541 shall be applied to the exterior and interior surfaces of the air cleaner. The filter element carrier and latch shall be cadmium plated in accordance with QQ-P-416, type XI, class 3 or zinc plated in accordance with ASTM-B633 type II SC-2. The zinc plating shall be a minimum of 0.3 roils thick. Stainless steel springs of the retainer assembly shall not require protective coating.

3.4 Performance.

3.4.1 Air leakage (see 4.6.1.1). The air cleaner, when subjected to a vacuum of 50 inches of water or a pressure of 2 pounds per square inch (psi) for a period of not less than 5 minutes, shall not leak more than 0.3 standard cubic feet per minute (scfm).

NOTE : This requirement does not apply to the filter element.

3.4.2 Airflow restriction (see 4.6.l.2). Pressure drop from intake to outlet shall not exceed 9.6 inches of water when filter is clean and airflow is 980 scfm.

3.4.3 Initial efficiency (4.6.l.3). Air cleaner initial efficiency shall be 99.5 percent minimum, as determined by the method specified in SAE J726 after 60 minutes of testing.

3.4.4 Dust capacity (see 4.6.1.4). Dust capacity shall be such that pressure drop across the air cleaner shall not exceed 20 inches of water after 28 hours of operation with a vacuum of 5.0 inches of water or a continuous air flow of 98 scfm at the scavenging outlet and dust continuously introduced at a rate of 0.025 gram per cubic foot (gcf) of air.

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3.4.5 Airflow restriction after cleaning (4.6.1.51. Pressure drop across the air cleaner shall not exceed 1.0 inch of water increase over original pressure drop after the air cleaner has been operated to dust capacity, filter element cleaned and the same filter element reinstalled.

3.4.6 Flame resistance (4.6.1.6). Filter element shall not continue to burn or smolder for more than 10 seconds after removal of an applied flame.

3.5 Environmental conditions.

3.5.1 Shock (see 4.6.2.1). The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to shock pulses of 40 gravity units (g) for 11.0 millisecond (ins) time duration applied along each of the three major axes.

3.5.2 Vibration (see 4.6.2.2). The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation after being subjected to vibration levels, in each of the three major axes, of 1.5 and 4.2 g in the frequency range of 5 to 500 to 5 Hertz (Hz) with a sweep time of 15 minutes.

3.5.3 Steam and waterjet cleaning (see 4.6.2.3). The air cleaner shall demonstrate no performance degradation and show no evidence of damage or deformation when subjected to steam cleaning using high pressure steam and cleaner conforming to P-D-220, followed by waterjet cleaning.

3.6 Nameplate and Product marking

3.6.1 Nameplate. Material and construction of nameplate shall conform to MIL-P-514, type III, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455. type 11, grade A, class 1. Nameplate shall be permanently attached with metal fastners, ‘located as shown on Drawing 19207-12307204.

NOTE : This requirement does not apply to the filter element.

3.6.2 Product marking. Unless otherwise specified in the contract or purchase order, the air cleaner shall be identified in accordance with MIL-STD-130 and Drawings 19207-12307204 and 19207-12329100.

3.7 Workmanship. Workmanship shall be of the quality necessary to provide air cleaners free from defects that would affect their performance and serviceability or present safety hazards to operating and maintenance personnel.

3.8 Interchangeability. To provide for interchangeability, the air cleaner shall conform to this specification and Drawings 19207-12307204 and 19207-12329100. All subassemblies/components having the same part number shall be functionally and dimensionally interchangeable.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perfom any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or-quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to determine conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Materials and manufacturing processes. The contractor shall maintain, and make available for Government review upon request, documentation verifying that materials and processes used in manufacture of the air cleaner conform to specified requirements. As a minimum, documentation shall confirm that materials are as specified in 3.2.

4.1.3 Procurement of filter elements. The following applies to the acquisition of replacement air filter elements (Drawing 19207-12329100) for use in the air cleaner assembly (Drawing 19207-12307204). Satisfactory performance of the air cleaner assembly is the result of a combined function of the housing/precleaned and the filter element, and the two are inseparable in meeting these requirements. As a method of qualifying filter elements alone, an air cleaner assembly must be used as a test housing. The housing must be thoroughly cleaned between individual filter element tests, and its continued integrity verified throughout the testing. All requirements specific to the filter element must be met as well as the performance criteria for the complete air cleaner assembly. In the process of qualifying filter elements, if the air cleaner assembly should fail testing, and that failure is attributable to multiple test exposures for the air cleaner housing, a new housing may be selected and the test rerun without penalty for the previous failure.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

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4.3 Test conditions and equipment.

4.3.1 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following standard (room) ambient conditions:

a. Temperature: 77 ± 18°F

b. Relative humidity: uncontrolled room ambient

c. Atmospheric pressure: site pressure

4.3.2 Test conditions.

4.3.2.1 General. Test specimens shall be stabilized at room temperature for 1 hour before tests are begun.

4.3.2.2 Temperature and humidity. All performance tests shall be conducted with air entering the air cleaner at a temperature of 75 ± 15°F and a relative humidity of 30 to 65 percent.

4.3.2.3 Dust feeding rate. When the feeding of dust is required as part of any test, the dust shall be fed at an average rate of 0.025 gram per cubic foot of air. The dust rate shall not vary more than plus or minus 0.005 gram per cubic foot during any part of the applicable test.

4.3.2.4 Test dust. Test dust shall be of two grades labeled fine and coarse. A typical chemical analysis is given in table 1 of SAE J726. Particle size distribution as measured with a Leeds and Northrup Microtrac Analyzer shall be per table 2A of SAE J726. Alternatively, particle size distribution as measured with a Coulter Counter Analyzer or a Roller Analyzer shall be per table 2B of SAE J726.

4.3.2.5 Airflow. Specified values of airflow are for standard atmospheric conditions of 29.92 inches of mercury barometric pressure and 80°F ambient temperature providing an air density of 0.073 pound per cubic foot (pcf). Actual airflow values during testing shall be corrected to standard conditions and shall be within ± 2 percent of specified values.

4.3.2.6 Airflow restriction. Actual airflow restriction values shall be corrected to airflow with a density of 0.073 pcf and shall be accurate to ± 0.1 inch of water.

4.3.2.7 Air cleaner mounting position. The air cleaner shall be mounted in its normal operating attitude and shall be electrically grounded (see figure 1).

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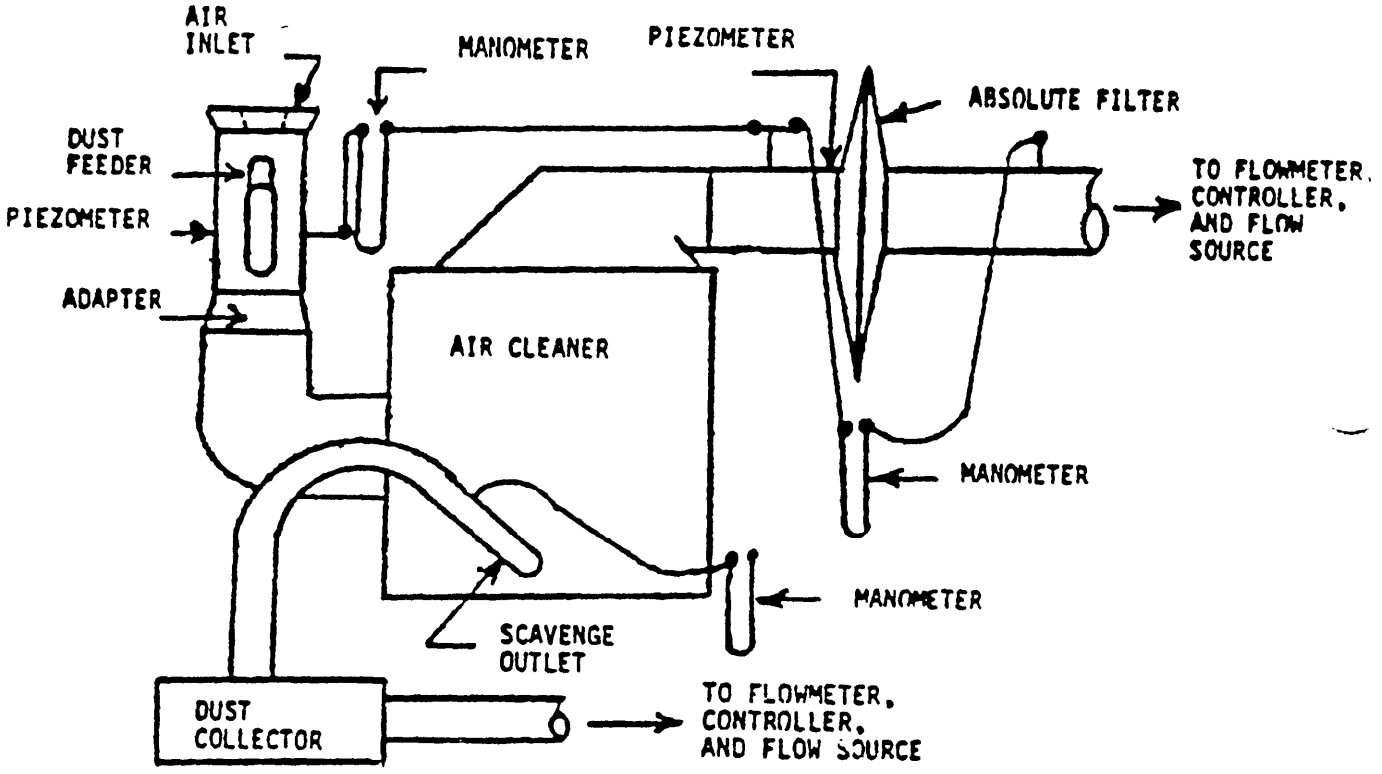


FIGURE 1. Test setup.

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4.3.3 Test equipment.

4.3.3.1 Piezometer tubes, transition ducts, adapter, manometer and flowmeter. Piezometer tubes conforming to figure 2 shall be installed at the inlet and outlet of the air cleaner. The inside diameter of the piezometer tubes shall be equal to the inside diameter of the air cleaner outlet (+0.5, -0.0 inch). Transition ducts shall be installed between the air cleaner inlet and inlet piezometer tube. Transition ducts and adapters shall have sides at angles no greater than seven degrees measured from the centerline of the duct or adapter. A smooth approach (figure 3) shall be installed at the front of the inlet piezometer tube. A water manometer shall be installed between the piezometer tubes (figure 1). A flowmeter shall be connected to the outlet of the absolute filter.

4.3.3.1.1 Test apparatus accuracy. Accuracy of test apparatus shall be as specified in MIL-ST&810 and SAE J726 as applicable.

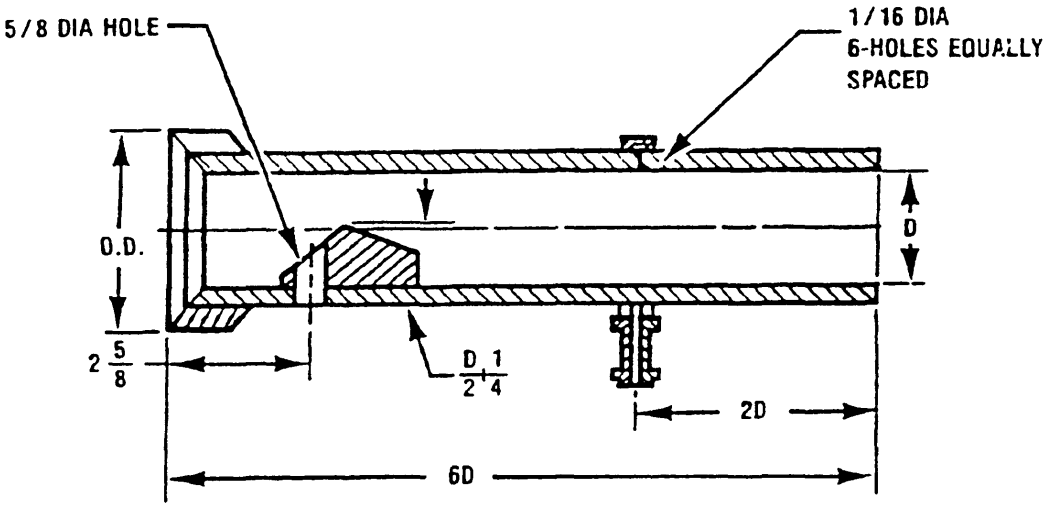
4.3.3.2 Absolute filter. The absolute filter shall consist of one layer of0.50-inch fiberglass batting conforming to SAE J7Z6 installed in a suitable housing with the nap side of the fiberglass batting facing upstream. The filter batting shall be of sufficient size to prevent excessive restriction. The face velocity of air against the filter batting shall not exceed 165 feet per minute. A manometer shall be located in the test setup to measure the pressure drop across the filter (see figure 1). A new filter batting shall be installed if, during the dust capacity test (3.4.4 and 4.6.1.4) the air pressure drop across the filter increases by 10 inches of water above its original restriction. The filter housing shall be electrically grounded during tests.

4.3.3.3 Drying and oven-weighing of absolute filter. The fiberglass batting shall be hung in an air oven and dried for 6 hours at a temperature of 200 to 240°F. Drying-time may be reduced, and the batting considered completely dry, if there is no change in the weight of the batting in two consecutive l0-minute intervals. All weighings shall be done with the fiberglass batting in the oven. The drying and overt-weighing procedure shall be observed before and after each test involving dust feeding.

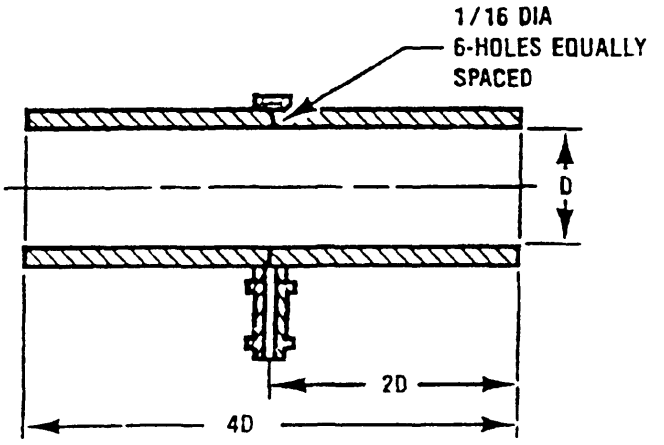
4.3.3.4 Scavenge vacuum. During all tests, except dust capacity test, a scavenge vacuum of 5.0 inches of water or a continuous air flow of 98 scfm shall be applied to the scavenge outlet. A piezometer tube of suitable capacity, a manometer and a dust collecting system shall be connected to ‘he scavenge outlet (see figure 1).

4.3.3.5 Dust feeder. The dust feeder (see figure 4) shall be of a type that will deliver dust to the cleaner without breaking down the dust particles or creating conglomerates. The feeder shall deliver the dust uniformly throughout the test. Dust at the beginning of testing shall have the same particle distribution as dust at the conclusion of testing.

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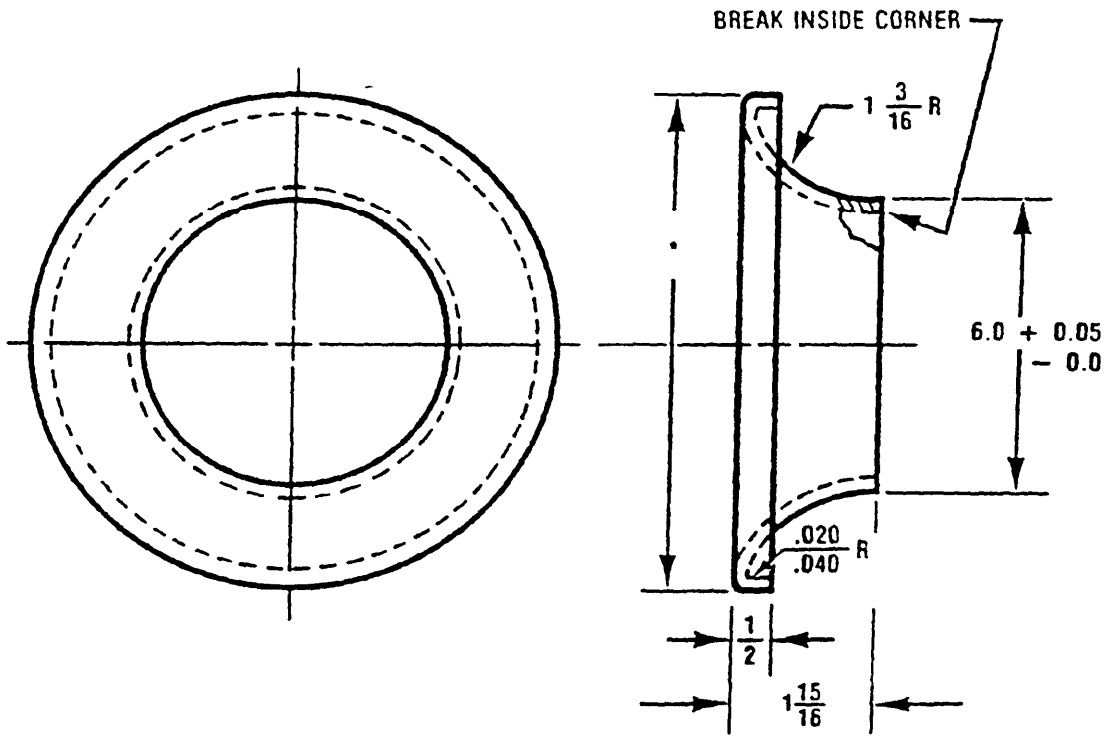
INLET PIEZOMETER TUBE



OUTLET PIEZOMETER TUBE

FIGURE 2. Piezometer tubes.

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* Diameter equal to O.D. of Inlet Piezometer (figure 2)

FIGURE 3. Ideal flow nozzle.

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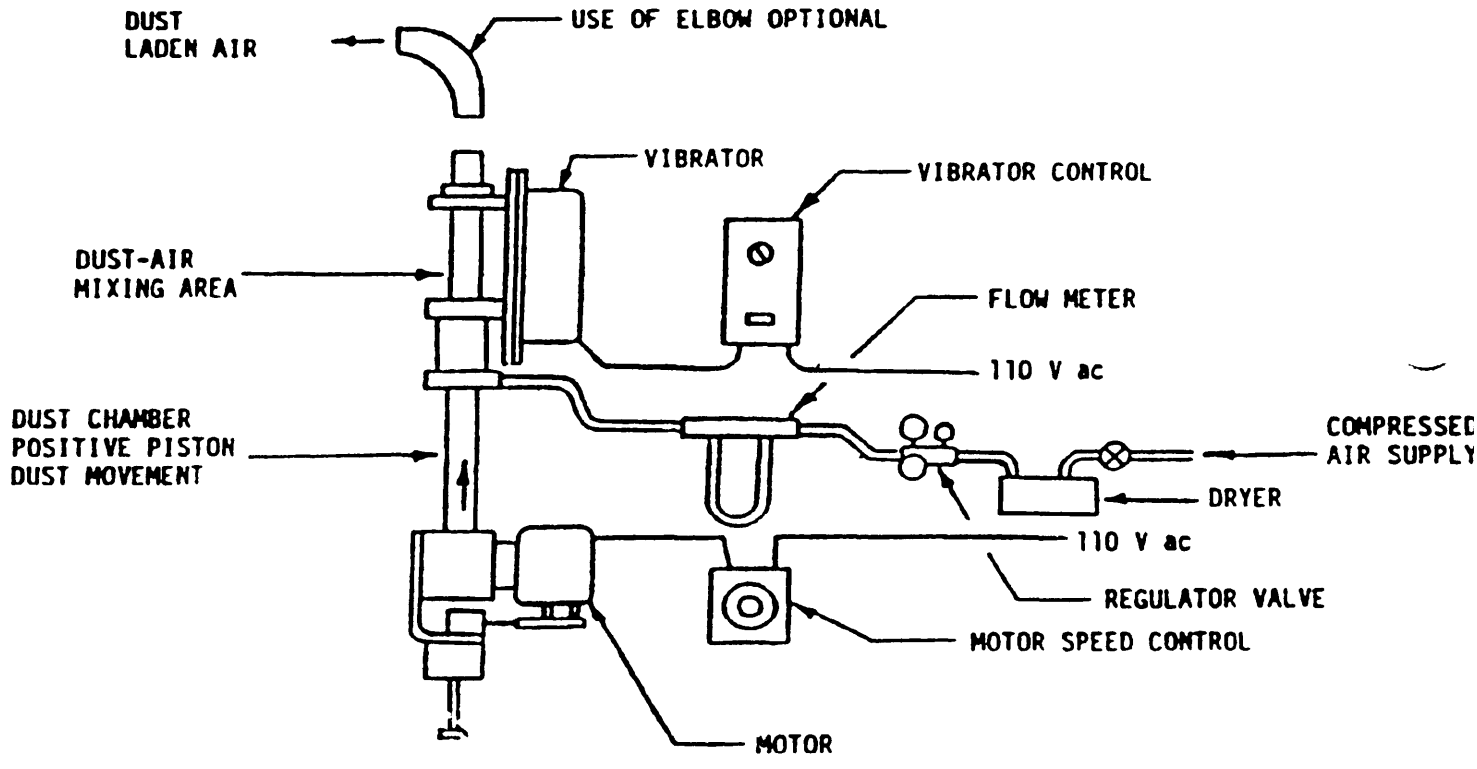


FIGURE 4. Dust feeder.

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4.4 First article inspection. When a first article sample is required (see 3.1 and 6.3), three air cleaners shall be subjected to first article inspection.

4.4.1 First article inspection routine. The first article samples shall be subjected to the inspections specified in table I.

TABLE I. First article inspection.

Inspection	Requirement Paragraph	Inspection Method
Group B	table III	table III
Weight (air cleaner)	3.3.1.1	Scale
Weight (filter element)	3.3.2.1	Scale
Air leakage	3.4.1	4.6.1.1
Airflow restriction	3.4.2	4.6.1.2
Dust capacity	3.4.4	4.6.1.4
Airflow restriction after cleaning	3.4.5	4.6.1.5
Flame resistance	3.4.6	4.6.1.6
Shock	3.5.1	4.6.2.1
Vibration	3.5.2	4.6.2.2
Steam and waterjet cleaning	3.5.3	4.6.2.3

4.4.2 First article failure. Inability of any of the first article samples to pass any examination or test shall constitute a failure. Failure of a sample to pass any examination or test shall be cause for refusal to grant first article approval. Such failure may, at the option of the Government, be cause for refusal to allow additional examinations or tests until causes of faults have been corrected and the corrective action approved by the Government,

4.4.3 Retention of first article approval. Unless extended by the Government to other contracts, first article approval is valid only on the contract under which it is granted.

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4.4.3.1 Changes to design, materials, processes, Procedures or location. Whenever a change is made in the design, materials, manufacturing processes or procedures, or facility location used in the production of air cleaners, the Government shall be notified to determine whether the contractor must requalify for first article approval.

4.4.4 Disposition of first article samples. Samples which have been subjected to first article inspection shall be indelibly marked "Test Equipment" and shall not be delivered as new equipment under the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections.

4.5.1.1 Group A inspection. Group A inspection shall consist of the test specified in table II.

4.5.1.1.1 Group A sampling 100 percent inspection shall apply to group A inspection.

TABLE II. Group A inspection.

Inspection	Requirement Paragraph	Inspection Method
Air leakage	3.4.1	4.6.1.1

4.5.1.2 Group B inspection. Group B inspection shall consist of the inspections specified in table 111.

TABLE III. Group B inspection.

Inspection	Requirement Paragraph	Inspection Method	Classification of characteristics	
			Major	Minor
Dimensions affecting mounting/interface	3.3.1.2 3.3.2.2	Standard Inspection Equipment(SIE)	x	
Dimensions not affecting mounting/interface	3.3.1.2 3.3.2.2	SIE		x
Protective coating	3.3.3	Visual		x
Nameplate and product marking	3.6	Visual		x
Workmanship	3.7	Visual		x

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4.5.1.2.1 Group B sampling. Frequency of inspection and/or any special arrangements for establishing the frequency of inspection shall be as specified in the contract or purchase order (see 6.2). Recommended requirements for a sampling plan are shown in 6.4.

4.5.2 Periodic inspection Periodic inspection shall consist of group C and D inspections as specified in tables IV and V. Delivery of air cleaners which have passed group A and B inspections shall not be delayed pending the results of group C and D inspections.

4.5.2.1 Group C sampling Unless othewise specified (see 6.2), sample air cleaners shall be selected at the rate of 1 per each 50 air cleaners produced or 1 every 3 months, whichever comes first.

4.5.2.2 Group D sampling Unless otherwise specified (see 6.2), sample air cleaners shall be selected at the rate of 1 per each 200 air cleaners produced or 1 every 6 months, whichever comes first.

4.5.2.3 Noncompliance. If a sample fails to pass either of the periodic inspections, the contractor shall notify the Government and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. After the corrective action has been taken, the periodic inspection shall be repeated on additional sample units (or the reworked failed unit(s) if so identified and agreed upon in the corrective action plan submitted by the contractor). All tests and examinations, or the test(s)/examination(s) which the original sample failed, shall be performed by the contractor at the option of the Government. Acceptance and shipment of product shall be discontinued until corrective action, acceptable to the Government, has been taken and the periodic inspection has shown that the corrective action was successful.

TABLE IV. Group C inspection.

Inspection	Requirement Paragraph	Inspection Method
Airflow restriction	3.4.2	4.6.1.2
Initial efficiency	3.4.3	4.6.1.3

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TABLE V. Group D inspection.

Inspection	Requirement Paragraph	Inspection Method
Airflow restriction	3.4.2	4.6.1.2
Initial efficiency	3.4.3	4.6.1.3
Dust capacity	3.4.4	4.6.1.4
Airflow restriction after cleaning	3.4.5	4.6.1.5

4.5.2.4 Disposition of samples. Sample air cleaners units which have been subjected to the periodic inspections may be delivered on the contract. The sample units shall be cleaned and new filter elements shall be installed. Sample filter elements shall not be delivered on the contract.

4.5.3 Nonconforming parts The contractor may rework or reprocess rejected parts to correct the nonconforming characteristic(s) and resubmit them for inspection. Reworking/reprocessing of parts shall be within the confines of drawing and specification requirements, and shall require 100 percent inspection.

4.5.4 Inspection of packaging The sampling and inspection of packaging shall be in accordance with 5.1.

4.6 Methods of inspection. The following methods of inspection are acceptable for verifying compliance with the requirements of 3.4 and 3.5.

4.6.1 performance tests.

4.6.1.1 Air leakage. To determine conformance to 3.4.1, the air cleaner inlet shall be connected to a flowmeter and all other inlets and outlets shall be sealed to prevent air leakage. The flowmeter shall be as accurate as a 3/16-inch orifice connected to a ± 0.01-inch water manometer. The flowmeter shall indicate an airflow no greater than 0.3 scfm when the air cleaner is subjected to a vacuum " of 50 inches of water when measured at the air cleaner air outlet. As an alternate method, air cleaner shall be pressurized at not less than 2 psi pressure; verify that air leakage does not exceed the specified limit.

4.6.1.2 Airflow restriction. To determine conformance to 3.4.2, the following test shall be conducted. With the air cleaner set up as shown in figure 1, airflow restriction in inches of water shall be determined at the rated airflow of 980 scfm and recorded.

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4.6.1.3 Initial efficiency. To determine conformance to 3.4.3, the following test shall be conducted. With the air cleaner set up as shown in figure 1, the airflow shall be adjusted to 980 scfm. Test dust in accordance with 4.3.2.4, fine grade, shall be fed in accordance with 4.3.2.3 for a period of 60 minutes with airflow restriction recorded at start of test, and at 10-minute intervals. The absolute filter shall be dried and oven-weighed in accordance with 4.3.2.3, 'taking care to retain all entrapped dust. Weighings shall be accurate to 0.01 gram, wherever possible, but never less accurate than 0.1 gram. Initial efficiency shall be determined by means of the following equation:

$$\text{Efficiency in percent} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where: W_1 is the weight of dust fed into air cleaner inlet.
W_2 is the weight of absolute filter and entrapped dust.
W_3 is weight of absolute filter before tests.

4.6.1.4 Dust capacity. To determine conformance to 3.4.4, the following dust capacity test shall be conducted. A new filter element shall be installed for this test. With the air cleaner set up as shown in figure 1, the airflow shall be adjusted to 980 scfm. Airflow restriction shall be recorded at the start of testing and at 10-minute intervals throughout the test. Test dust in accordance with 4.3.2.4, coarse grade, shall be fed in accordance with 4.3.2.3. After obtaining the initial airflow restriction at rated airflow, the airflow shall be varied as shown in figure 5. Figure 5 is based on SAE J726 and will result in an average airflow which is 60 percent of rated airflow. This airflow cycle shall be continued for the duration of this test. The dust capacity test shall be terminated when airflow restriction exceeds 20 inches of water, or at the end of 28 hours, whichever occurs first.

4.6.1.5 Airflow restriction after cleaning. To determine conformance to 3.4.5, after the dust capacity test (4.6.1.4) the filter element shall be removed from the air cleaner and compressed-air-cleaned as follows. The compressed air shall be applied first in a direction opposite to normal airflow, and then in the direction of normal airflow. Line air pressure shall be 100 ± 10 psig, and the nozzle inside diameter shall be nominal 0.12 inch. After the filter element is thoroughly cleaned, it shall be reinstalled in the air cleaner. Adjust the airflow to 980 scfm and determine the airflow restriction.

4.6.1.6 Flame resistance. To determine conformance to 3.4.6, expose a sample of the filter medium used in the filter element to a flame until the sample burns or glows. Remove the flame and observe.

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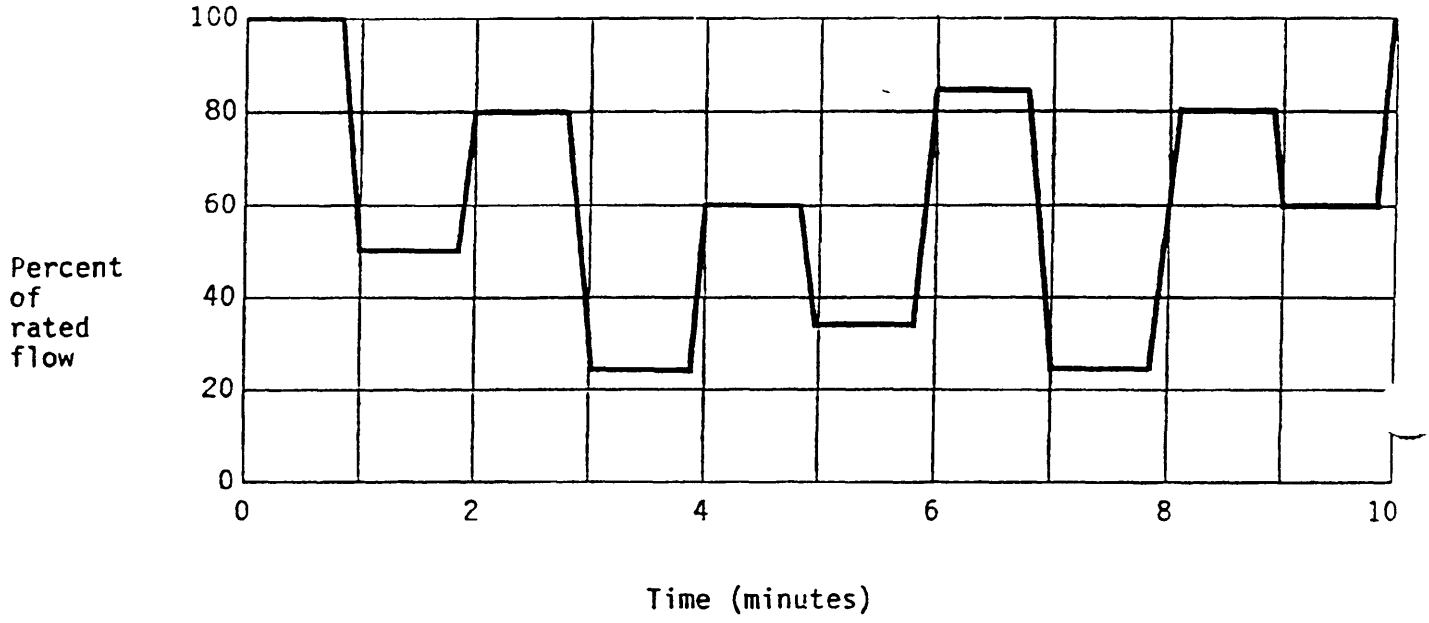


FIGURE 5. Airflow variation.

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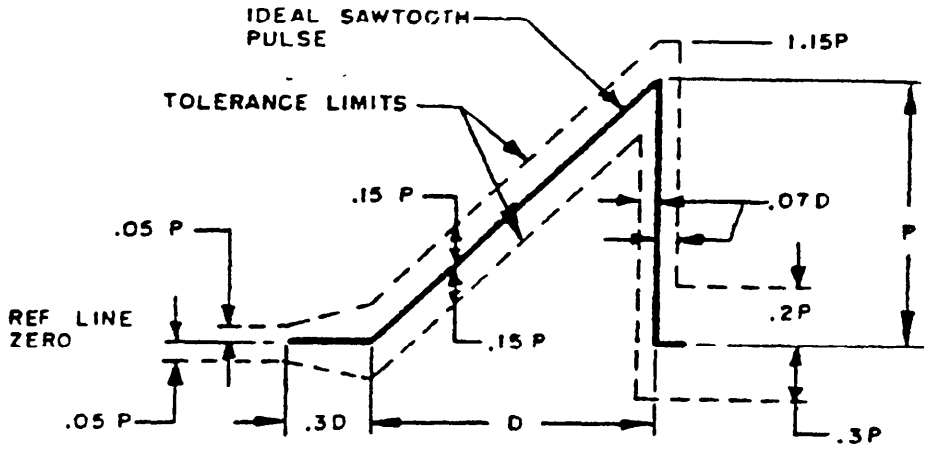
4.6.2 Environmental tests.

4.6.2.1 Shock. To determine conformance to 3.5.1, the air cleaner shall be mounted in a manner that simulates actual use, or shall be rigidly mounted to a shock table, and subjected to three sawtooth shocks of 40g and time duration of 11 ms (see figure 6) along the three mutually perpendicular axes of the air cleaner for a total of 18 shocks. No performance testing shall be required during shock. At the conclusion of shock testing, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.6.1.1 and 4.6.1.3.

4.6.2.2 Vibration. To determine conformance to 3.5.2, the air cleaner shall be mounted, in a manner that will simulate its actual use, and vibrated along each axis in accordance with the vibration levels specified in figure 7. The sweep time shall be 15 minutes for the sweep frequency range of 5 to 500 to 5 Hz. The cycling time shall not exceed 3 hours in each axis. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 8. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown in figure 8 for the specified range. When the air cleaner resonances below 5 Hz are measured, the test curve shall be extended to 2 Hz and the sweep time shall be 18 minutes (2 to 500 to 2 Hz). There shall be no operational tests required during vibration. After the test, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.6.1.1 and 4.6.1.3.

4.6.2.3 Steam and waterjet cleaning To determine conformance to 3.5.3, with the inlet and outlet tubes suitably sealed, the air cleaner shall be subjected to steam cleaning, followed immediately by waterjet cleaning as follows. The jet shall be applied perpendicular to the surface being cleaned at a distance of not greater than 1 foot for steam cleaning, and not greater than 3 feet for waterjet cleaning. The jet pressure shall be 105 $\pm$ 5 pound-force per square inch gage (psig) for steam cleaning and 50 + 5 psig for waterjet cleaning. The jet shall be moved at such a rate that the exposed surface of the air cleaner is subjected to the jet at the rate of 1 square foot per minute for 2 to 3 minutes. Prior to and within 2 hours after completing the steam and waterjet cleaning, the air cleaner shall be subjected to the air leakage and initial efficiency tests specified in 4.6.1.1 and 4.6.1.3. -Moisture inside the air cleaner shall not be considered a failure

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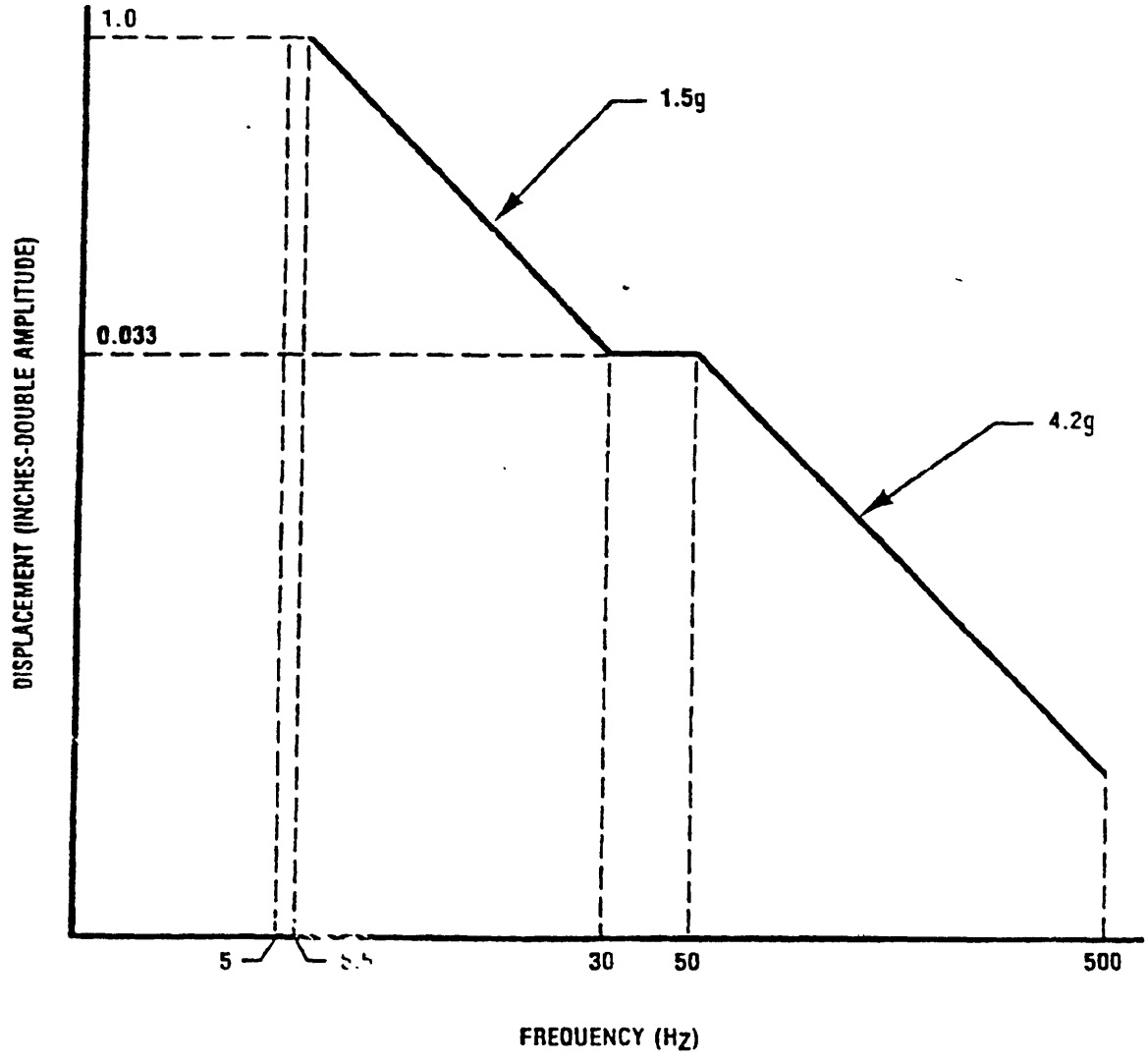


P = Peak value = 40 g
D = Nominal duration = 11ms

NOTE : The oscillogram shall include a time about 3D long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries, and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi $\pm$ 0.1 V_j, where V_j is the velocity-change associated with the ideal pulse which equals 0.5 PO. The integration to determine velocity change shall extend from 0.40 before the pulse to O.lD after the pulse.

FIGURE 6. Terminal-peak sawtooth shock Pulse configuration and tolerance limits.

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NOTES :

1. Acceleration levels: ± 9 g (peak).
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 7. Vibration test curve.

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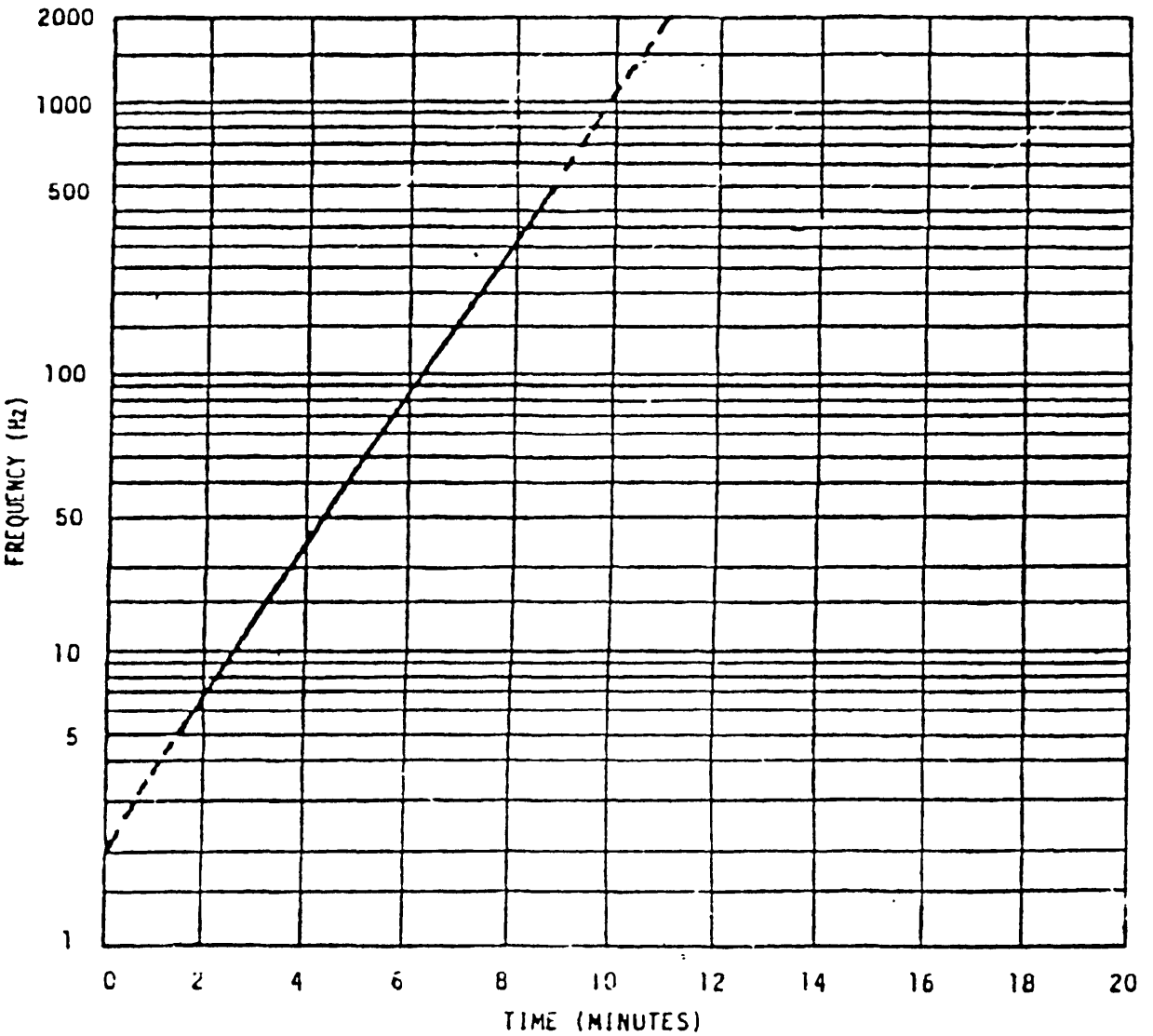


FIGURE 8. Logarithmic sweep.

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Air cleaners manufactured in accordance with this specification are intended for use on military tracked vehicles in the 500 HP AO, Al and AZ Bradley Fighting Vehicle (BFV) Family.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision, and date of this specification.

b. Title, number, revision, and date of part drawing.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

d. First article inspection, if required (see 3.1).

e. Responsibility for inspection, if other than as specified herein (see 4.1).

f. Sampling plan requirements for group B inspection (see 4.5.1.2.1).

9* Periodic sampling plan if other than as specified herein (see 4.5.2.1 and 4.5.2.2).

h. Test sample disposition, if other than as specified herein (see 4.5.2.4).

i. Selection of applicable level and packaging requirements (see 5.1).

j. Any deviation from this specification.

6.3 First article inspection. When first article inspection is required (see 3.1), the contracting officer should provide specific guidance to bidders whether it should include preproduction samples and/or initial production samples, or standard production samples from the contractor's current inventory. The acquisition document should include specific instructions regarding the arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive first article inspection, or any portion thereof, to those bidders offering a product which has been previously acquired or tested by the Government or other responsible testing activity. Bidders offering such products should furnish evidence with the bid that prior first article approval is presently appropriate for the pending contract, or that first article inspection or any portion thereof, is inappropriate or unnecessary for the pending contract.

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6.3.1 Qualified source. A contractor complying with requirements of first article inspection may be identified on source control Drawings 19207-12307204 and 19207-12329100 as a-qualified source. The Bradley Fighting-Vehicle System Program Management Office (BFVS PMO) is responsible for maintaining records of qualified sources.

6.4 Recommended group 3 sampling plan (see 4.5.1.2.1).

6.4.1 Basic Method. Quality characteristics listed in table 111 must be inspected using the sampling plan shown in table VI. Alternate sampling plans are not allowed except under an approved statistical process control (SPC) system conforming to the requirements outlined in paragraph 6.4.2.

TABLE VI. Group B sampling plan.

Lot Size	Sample Size	
	For Major Characteristics	For Minor Characteristics
2 to 13	Entire lot	3
14 to 25	13	3
26 to 50	13	5
51 to 90	13	6
91 to 150	13	7
151 to 280	20	10
281 to 500	29	11
501 to 1200	34	15
1201 to 3200	42	18
3201 to 10,000 ‘	50	22

NOTE : Accept lot on zero defects.
Reject lot on one or more defects.

6.4.1.1 lighteded inspection. Tightened inspection must be introduced as soon as 2 out of 5 successive lots (see 6.4.1.3) have been rejected and must as a minimum, impose a 30 percent increase in sample size. Normal inspection sampling may be restored after 5 successive lots have been accepted under tightened inspection.

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6.4.1.2 Reduced inspection. Reduced inspection may be introduced when 10 successive lots have been accepted and must as a maximum, permit a 30 percent decrease in sample size. Normal inspection sampling must be restored if a lot is not accepted under reduced inspection.

6.4.1.3 Lot. A lot is defined as a collection of parts produced under essentially the same conditions and offered for inspection at one time.

6.4.1.4 Rejected lot. A rejected lot may be 100 percent sorted (screened) to remove nonconforming parts and to determine acceptance of conforming parts.

6.4.2 Statistical Process control (SpC). Sampling plan requirements may be satisfied through the use of SPC as an alternate control method. If SPC is to be used, an SPC plan, detailing the methods for monitoring process control, must be approved by the Government or agent. The plan, developed by the contractor, must include as a minimum:

a. Types of control charts and their uses.

b. Process capability (C_{pk}) studies.

c. Criteria for determining out-of-control conditions.

d. Corrective actions to be taken if an out-of-control or out-of-tolerance condition is detected.

e. Training programs and qualification requirements of personnel executing the plan.

f. Provisions for reduced inspection.

The plan must contain the results of process capability studies for the processes being controlled. For variable data, the minimum C_{pk} for statistical control must be 1.33. For attribute data, a minimum process average of 99.73% is required

6.5 Definitions.

6.5.1 Air cleaner. The air cleaner is a sheet metal enclosure approximately 38 inches wide, 45 inches high and 10 inches deep. The unit is a multistage system composed of an-integral first stage inertial separator, or precleaned, and-a second stage replaceable barrier filter element. The air cleaner has a tubular inlet, a primary gravity precipitator with scavenging exhaust outlet, a replaceable filter element and a tubular outlet.

6.5.2 Filter element. The filter element is an easily replaceable component essential to the overall performance of the air cleaner assembly. The filter element is reusable and capable of multiple cleanings.

6.6 Supersession data. This specification supersedes MIL-A-62565(AT), dated 22 June 1989, and FMC Drawing 19207-12292158, dated 26 October 1992.

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6.7 Subject term (key word) listing.

Absolute filter
Air leakage
Airflow
Dust
Flame resistance
Piezometer
Scavenge vacuum

6.8 Changes from Previous issue. Marginal notations are not used in this revision to identify changes with respect tc the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2940 -A017)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-62565A(AT)	2. DOCUMENT DATE (YYMMDD) 940415

3. DOCUMENT TITLE
Air Cleaner, Engine

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (810) 574-5954 (2) AUTOVON: 786-5954
c. ADDRESS (Include Zip Code) U.S. ARMY TANK-AUTOMOTIVE COMMAND ATTN: AMSTA-GDS WARREN, MICHIGAN 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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